QUALITY ASSESSMENT REPORT

1. Cover Information

Evaluator Name:	Hamza Kazmi , Fahad Shafique
Student ID:	FA23-BSE-063, 021
Section / Group:	A
Course:	Software Quality Engineering — SQE Lab
Instructor:	Engr. Asim Javaid
App 1:	YouTube Music *(v9.16.4)*
App 2:	Spotify *(v9.1.42)*
Device Model:	Iphone 16e
OS Version:	26.3.1
Network Conditions:	Campus WiFi / 4G
Evaluation Date:	26 April 2026
Pair Letter:	E

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2. App Overviews

YouTube Music is a music streaming application developed by Google, designed to provide users with access to a wide range of songs, albums, and music videos. It targets users who prefer both audio and video-based music content. Key features include personalized recommendations, playlists, and integration with YouTube content. Compared to other apps, it offers more flexibility in discovering unofficial or remix content.

Spotify is a widely used music streaming platform focused on audio-based streaming. It targets a broad audience with features such as curated playlists, podcasts, and strong personalization. Spotify provides a more structured and polished experience compared to YouTube Music, with better playlist management and cross-device integration.

Both apps serve the same primary purpose (music streaming), but differ in **content style (video vs audio)** and **user experience design**.

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3. Evaluation Methodology

The evaluation followed the **ISO 25010 quality model**, using a **1–5 Likert scale** where:

- 1 = Poor
- 3 = average
- 5 = Excellent

Each app was tested under the same conditions:

- Device: iPhone
- OS: iOS 26.3.1
- Network: Same WiFi/4G conditions
- Duration: ~10–15 minutes per characteristic

Testing methods included:

- Stopwatch for timing performance
- Manual interaction for usability
- Settings inspection for security and permissions
- App Store review for maintainability
- Airplane mode and force-close tests for reliability

Screenshots were taken as evidence for each test case.

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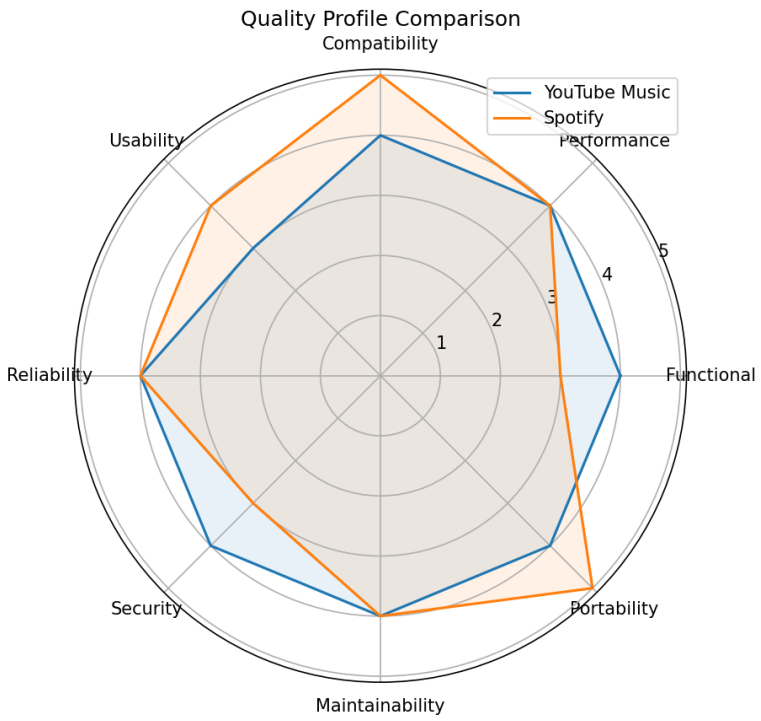
4. Completed Checklists (Summary)

COMPARATIVE RATING SUMMARY TABLE

Quality Characteristic	Weight	App 1 Rating	App 2 Rating	Gap (|A1–A2|)	Better App
1. Functional Suitability	15%	4	3	1	App 1
2. Performance Efficiency	12%	4	4	0	Tied
3. Compatibility	10%	4	5	1	App 2
4. Usability	15%	3	4	1	App 2
5. Reliability	15%	4	4	0	Tied
6. Security	15%	4	3	1	App 1
7. Maintainability	8%	4	4	0	Tied
8. Portability	10%	4	5	1	App 2
		App 1 Weighted	App 2 Weighted	Margin	
WEIGHTED TOTAL SCORE		3.85	3.90	0.05	App 2 — youtube music

5. Quality Profile Chart



6. Comparative Analysis

Overall, Spotify achieves a higher quality score compared to YouTube Music due to its stronger performance in usability, compatibility, and portability. While both applications perform well in core music streaming functionality, their differences become more noticeable in user experience and platform support.

One major quality gap is in **usability**. Spotify provides a cleaner and more intuitive interface, allowing users to complete tasks faster with fewer errors. In contrast, YouTube Music has slightly more complex navigation, especially when accessing settings or help features.

Another key difference lies in **compatibility and portability**. Spotify supports a wider range of platforms, including desktop applications for Windows and macOS, while YouTube Music is mainly limited to mobile and web. This gives Spotify an advantage for users who switch between devices frequently.

A trade-off exists in **security**. YouTube Music benefits from Google's account system, which includes strong two-factor authentication, making it more secure. Spotify, on the other hand, lacks consistent 2FA support, reducing its security rating.

In terms of **performance**, both apps perform well, but Spotify is slightly more efficient due to its audio-focused design, while YouTube Music may consume more data and battery due to video integration.

Overall, Spotify is recommended for users who value **performance, usability, and cross-platform access**, while YouTube Music is better suited for users who prefer **video-based content and stronger account security**.

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7. Recommendations

YouTube Music

App 1 — Recommendation 1	
Field	**Your Answer**
App Name	YouTube Music
Specific Action	Add offline download feature for free users (limited version
Characteristic Impr	Functional Suitability
Current Rating	4
Target Rating	5
Specific Deficiency	Free users cannot download music for offline listening
User Benefit	Users can listen without internet, improving usability and experience
Complete Recomm	Introduce a limited offline download feature for free users (e.g., low-quality downloads or limited songs) to improve accessibility and overall functionality.

App 1 — Recommendation 2	
Field	**Your Answer**
App Name	YouTube Music
Specific Action	Improve navigation and settings layout
Characteristic Impr	Usability
Current Rating	3
Target Rating	4
Specific Deficiency	Settings and help sections are hard to find
User Benefit	Easier navigation and faster task completion
Complete Recomm	Redesign the interface to make settings, help, and key features more visible and easier to access for users.

Spotify

App 2 — Recommendation 1	
Field	**Your Answer**
App Name	Spotify
Specific Action	Add full 2FA security option for all users
Characteristic Impr	Security
Current Rating	3
Target Rating	5
Specific Deficiency	Limited or no built-in two-factor authentication
User Benefit	Better account protection and data security
Complete Recomm	Implement a strong built-in two-factor authentication system (OTP or authenticator apps) to improve account security

App 2 — Recommendation 2	
Field	**Your Answer**
App Name	Spotify
Specific Action	Allow more playback control in free version
Characteristic Impr	Functional Suitability
Current Rating	3
Target Rating	4
Specific Deficiency	Free users are restricted to shuffle playback and ads
User Benefit	Better listening experience and user satisfaction
Complete Recomm	Provide limited song selection control (e.g., a few skips or direct play options) for free users to improve usability without affecting premium model.